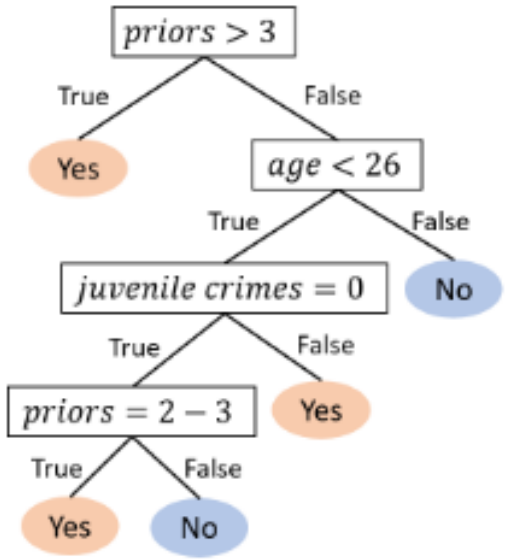
Machine Learning Fundamentals

Logical Models

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Logical Models



if $(age < 26)$ **and** $(priors = 2-3)$ **then predict** yes
else if $(juvenile\ crimes = 0)$ **and** $(priors < 3)$ **then predict** no
else predict yes

(b)

if $(priors > 3)$ **and** $(age < 21)$ **then predict** yes
if $(juvenile\ crimes > 0)$ **and** $(prior > 3)$ **then predict** yes
if $(age < 23)$ **and** $(prior = 2-3)$ **then predict** yes
else predict no

- Decision trees: Each path to a leaf is the antecedent of a rule.
- Decision lists: The rules are tried in order.
- Decision sets: Unordered collection of rules.

Logical Models

- Models composed of "if. . . then. . . " rules.
- Human-understandable: both models and individual predictions.

Can handle

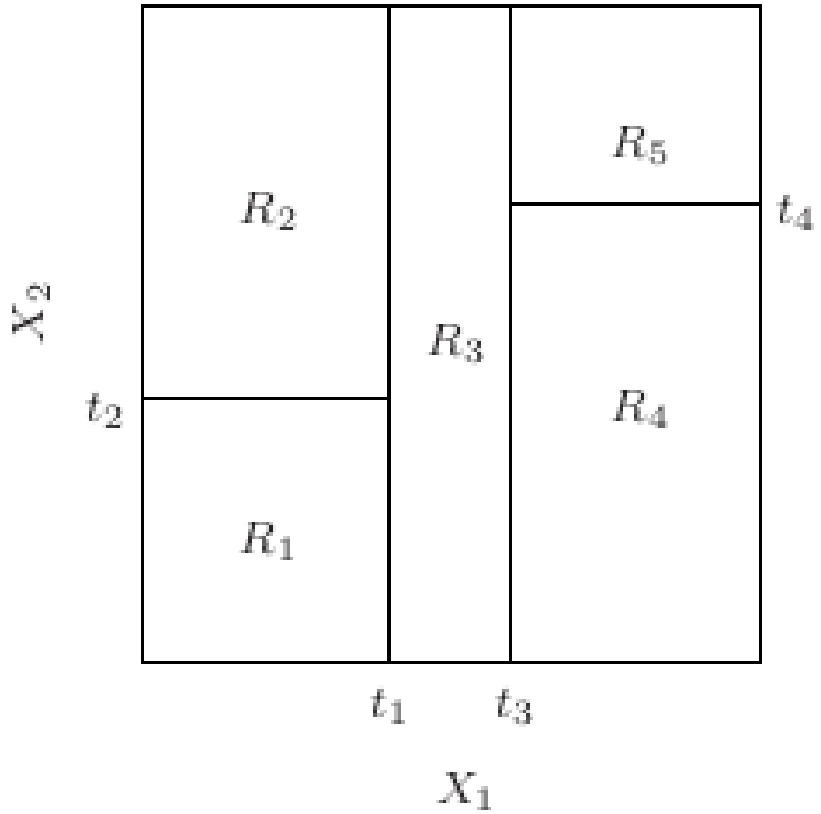
- Complicated interactions.
- Nonlinear effects.
- Outliers.

Example interaction: Gender and blood pressure

```
IF (gender = female AND blood pressure = high
    THEN predict Condition = TRUE".
IF (gender = male AND blood pressure = low)
    THEN predict Condition = FALSE".
```

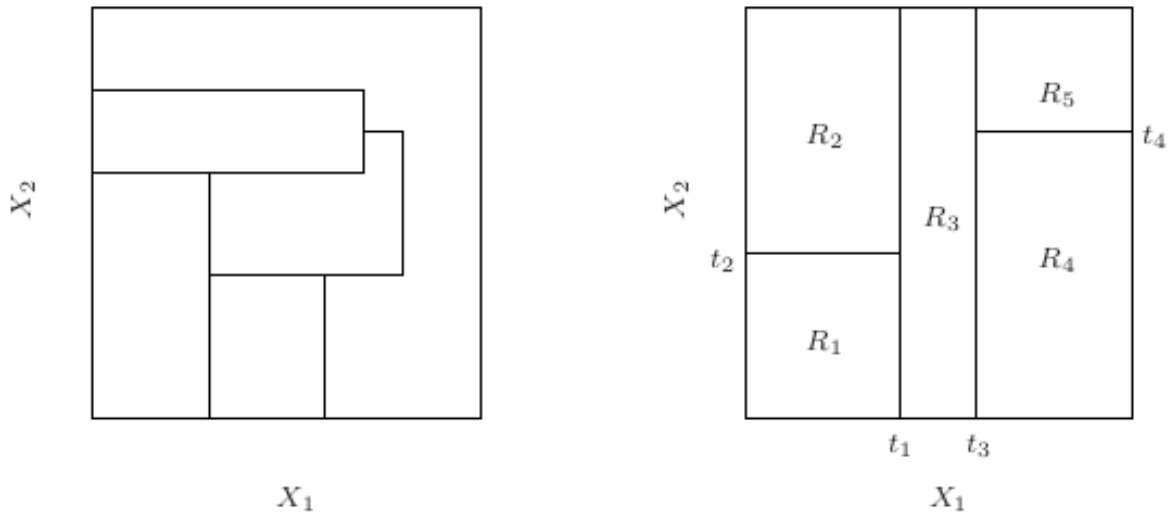
Trees

- Partition the feature space into rectangles.
- Partition by recursive binary partitions (as if cutting a paper in two).
- Example: $R_3 = \{X \mid t_1 \leq X_1 \leq t_3\}$.



Trees

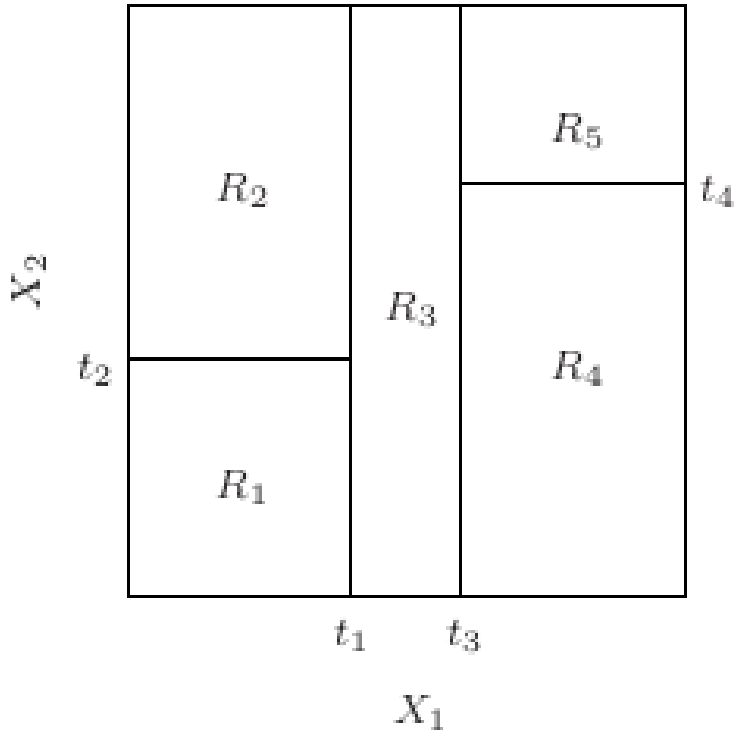
- A tree is restricted to regions such as those on the right figure.

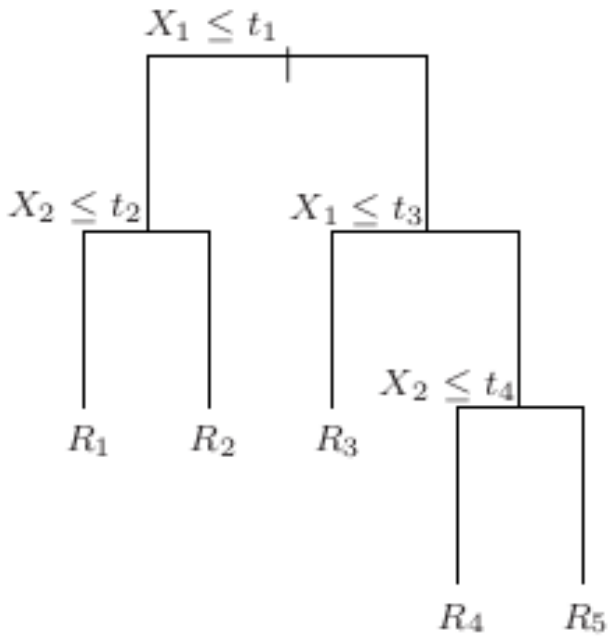


- Although the regions in the left figure are built with simple, univariate conditions (e.g., $X_1 > t$), they cannot be obtained by binary splits.

Trees

- The recursive, hierarchical splitting can be visualized as a binary tree.
- The regions R_m correspond to the leaves of the tree.



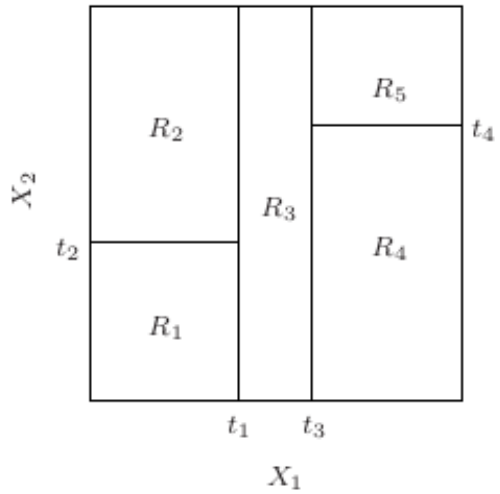


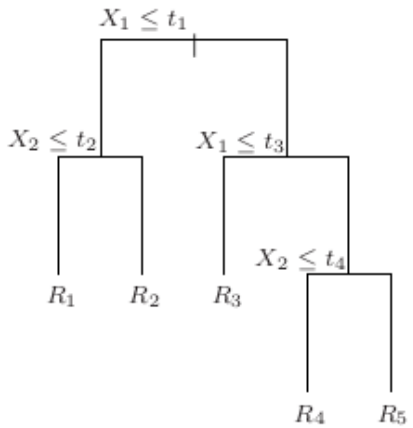
Trees

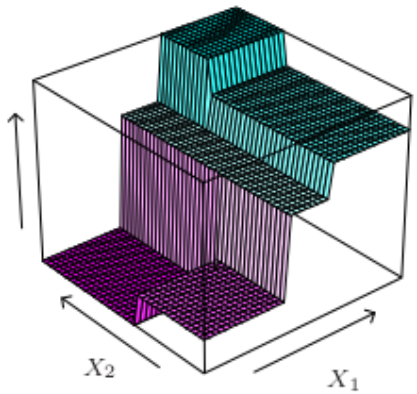
- The prediction is a constant in each region,

$$f(x) = \sum_{m=1}^{M} c_m I\left(x \in R_m\right).$$

Example: Regression with two quantitative features



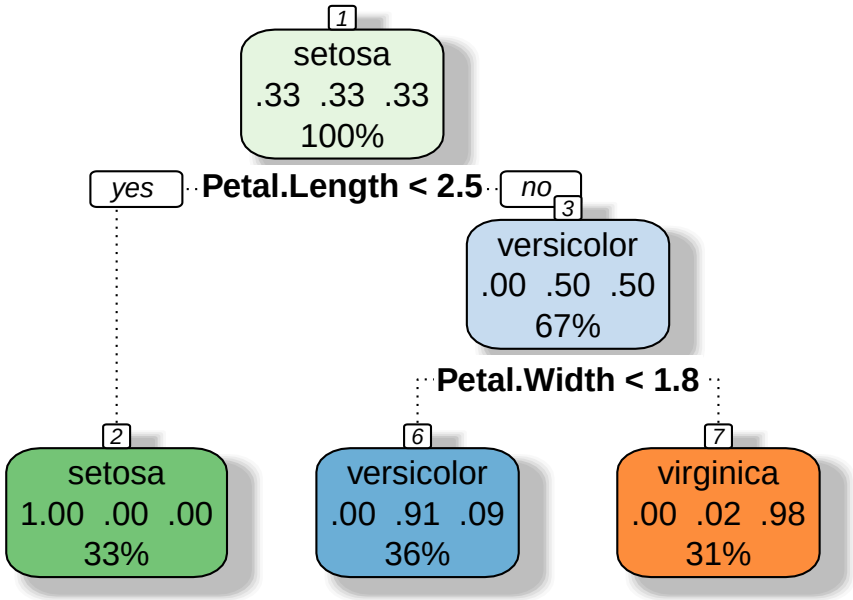




Interpretation

Binary tree visualization

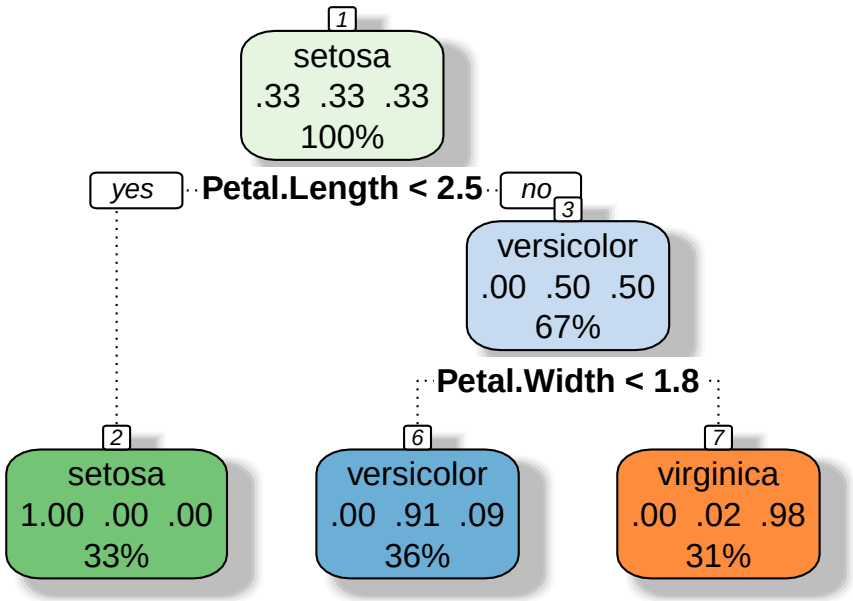
- While partitions are difficult to draw with $n > 2$, the binary tree representation works regardless of dimension.
- Not all features necessarily appear in the tree, and the tree can be sparse even in a high-dimensional setting.
- This tree uses only two out of four features in the Iris dataset.



Interpretation

Binary tree visualization

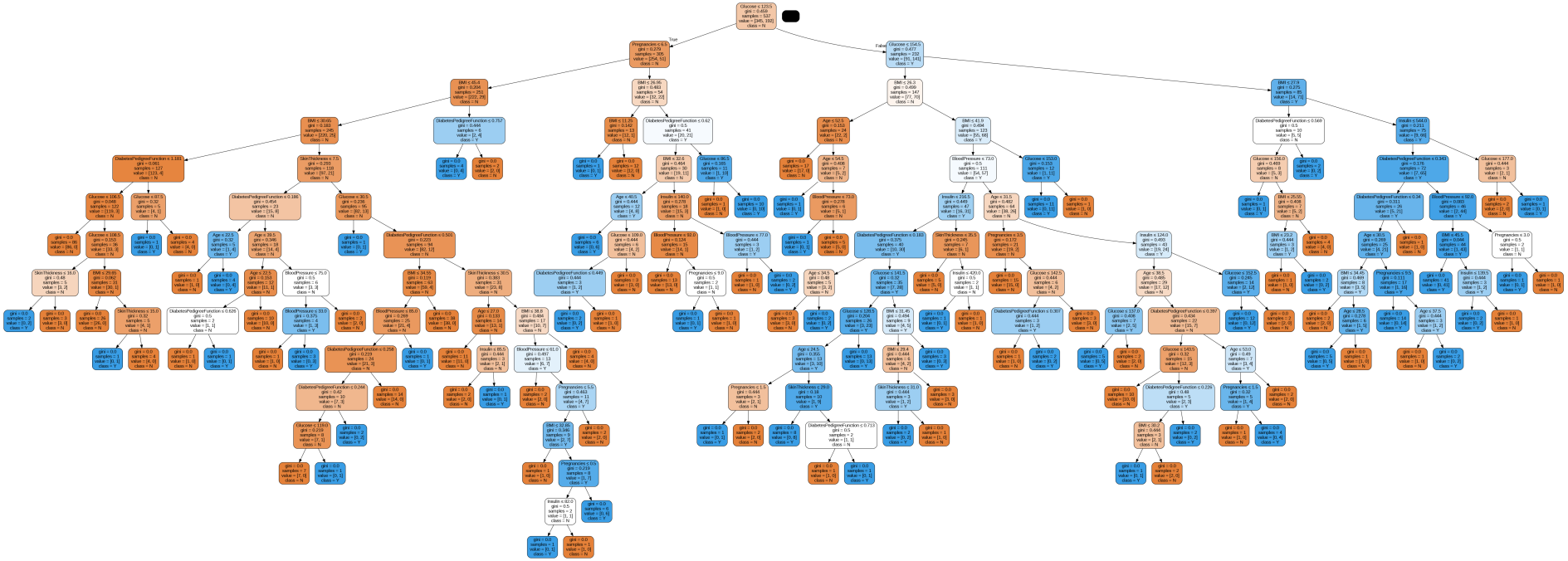
- We often display the number (proportion) of training set instances that reach the leaf (i.e., the support of the rule)
- As well as the training set class proportions in the leaves.



Interpretation

Binary tree visualization

A tree that is not sparse is not simulatable.

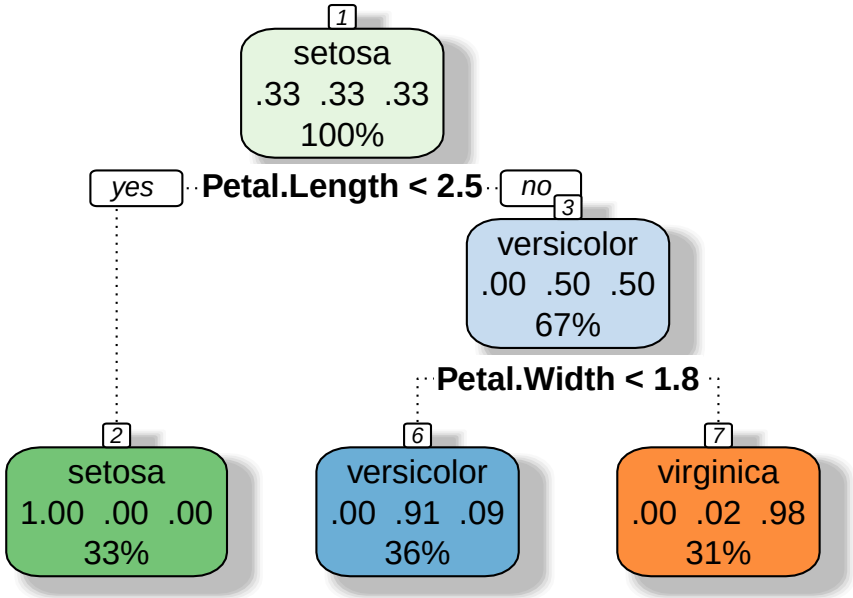


However, each of its predictions might be – local interpretability.

Interpretation

Extracting rules from a tree

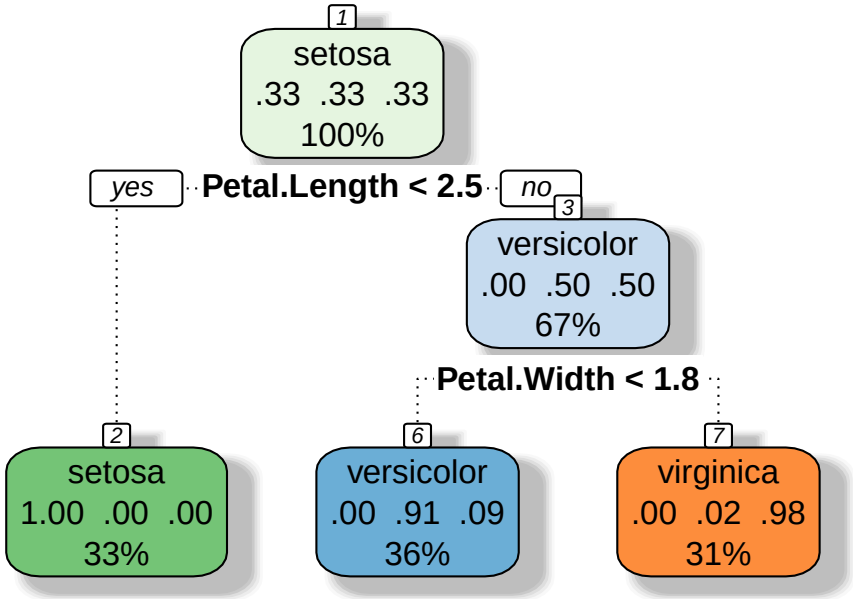
- A rule per leaf R_m: The antecedent is membership in the leaf and the consequent is the corresponding prediction: $x \in R_m \Rightarrow$ predict c_m.
- Specify the antecedent, $x \in R_m$, by, starting from the root, traversing nodes and connecting them with an "AND", until reaching the leaf.



Interpretation

Extracting rules from a tree

- IF petal length $<$ 2.5 THEN predict setosa.
- IF petal length $\geq$ 2.5 AND petal width $<$ 1.8 THEN predict versicolor.
- IF petal length $\geq$ 2.5 AND petal width $\geq$ 1.8 THEN predict virginica.



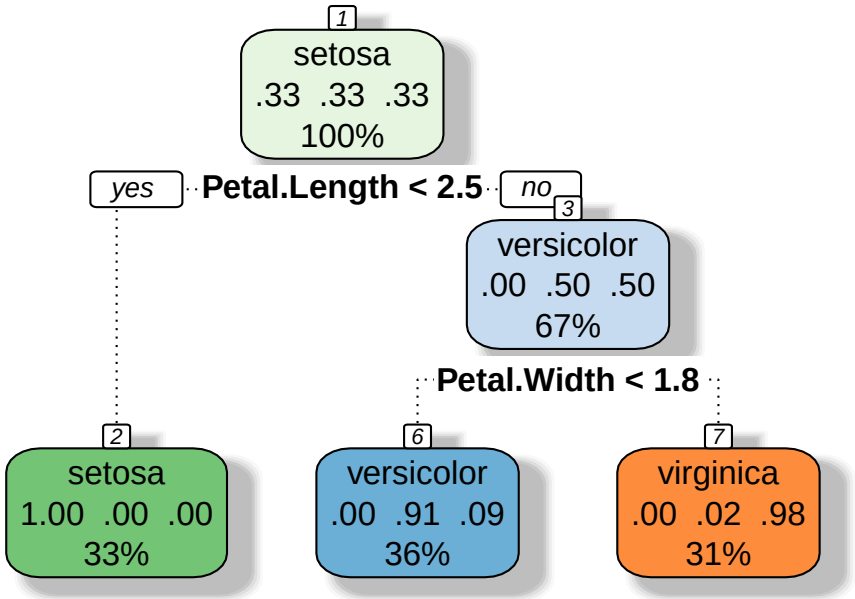
Interpretation

Extracting rules from a tree

IF petal length ≥ 2.5 AND petal width < 1.8 THEN predict versicolor.

Can also output class probabilities:

IF petal length ≥ 2.5 AND petal width < 1.8 THEN probability of versicolor is 91%.



Local interpretation

- The instance is 'sent' down the tree until reaching a leaf.
- Its prediction is due to the sole rule given by its path down to the leaf.

Explanations

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- Selected: as long as the tree is very shallow: depth-three tree explains each prediction with at most thee splits.
- Truthful: as long the tree is accurate.
- Note that trees are rarely very deep because the number of leaves is exponential in depth.

Learning

Once we have the regions R_m, estimating the constants c_m is easy.

- Regression: typically, c_m = mean of training observations in the R_m, as that minimizes squared error.
- Classification: typically c_m = the mode, as it minimizes 0/1 loss.

Computational complexity

- Identifying the optimal binary partition in terms of a score such as squared error is generally unfeasible.
- Indeed, decision tree optimization is an NP-complete problem.
- Thus, learning typically tackled with greedy growing and pruning algorithms.

Learning

Greedy tree learning pseudo-algorithm (regression, quantitative features)

1. Consider a splitting variable j and split point s, and define the pair of half-planes $R_1(j,s) = \{X \mid X_j \leq s\}$ and $R_2(j,s) = \{X \mid X_j > s\}$. Seek the splitting variable j and split point s that solve

$$\min_{j,s} \left[\min_{c_1} \sum_{x_i \in R_1(j,s)} (y_i - c_1)^2 + \min_{c_2} \sum_{x_i \in R_2(j,s)} (y_i - c_2)^2 \right].$$

   For a fixed X_j, s, c_1, and c_2 can be found rather quickly.
2. Having found the best (j,s), partition the data into the two resulting regions and repeat the splitting process (step 1) for each of the two regions, until some stopping criterion.

Learning

A large/small tree may overfit/underfit the data.

Tree size

- Controlled by parameters as maximum depth, minimum leaf size, etc.
- Typically estimated from data, rather than pre-specified.

Post-pruning

- Grow a large tree;
- then prune it by balancing prediction and tree size.

Advantages

- Insensitive to monotonic transformations of inputs, such as logarithms, and thus often requires no preprocessing.
- Can combine categorical and continuous features.
- Can express interactions, non-linear, and non-monotonic relation between the features and the outcome.
- Insensitive to the presence of many irrelevant variables.

Disadvantages

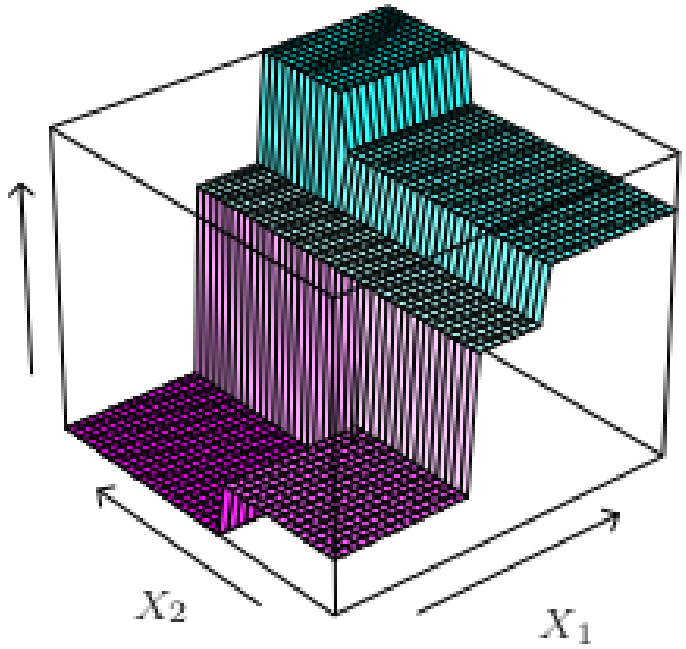
Difficulty in capturing additive or linear structure

- Additive relationships need be approximated by axis-parallel rectangles.
- Let $Y = c_1 I(X_1 < t_1) + c_2 I(X_2 < t_2) + \varepsilon$, with $\varepsilon \sim \mathcal{N}(0, 1)$.
- A binary tree might make its first split on X_1 near t_1.
- Then, it would need to split both nodes on X_2 at t_2; may not happen if, e.g., insufficient data.
- Many such additive effects would yield a complex tree.
- No single coefficient to quantify the additive effect of a feature.

Disadvantages

Lack of smoothness

Slight changes in the input feature can have a big impact on the predicted outcome; especially problematic in regression.



Disadvantages

High-variance / instability

- Often a small change in the data can result in a very different trees.
- The major reason for instability is the hierarchical nature of the model: if a different feature is selected for the first split, the entire tree changes.
- A major reason why we may select, under small perturbations, different initial splits is the greedy nature of the learning algorithms.

Rashomon effect

- Possibly multiple accurate yet very different trees for a single problem.
- This reduces confidence in any single tree being the "true" model.
- Variable importance: actually informative features may not be selected by an accurate tree, due to redundancies.

Instability

Discussion

- As we have noted, instability may reduce confidence in any single model being the "true" model.

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Discussion

- As we have noted, instability may reduce confidence in any single model being the "true" model.
- However, we are often not aiming to discover the "true" model, but rather just a useful model.
- In this case, what we care about is that the model is transparent and that we can understand its predictions, regardless of our confidence in it being the "true" model.

Predictive performance

Limited accuracy

- The main limitation however is the, often, limited predictive power.
- This is mainly a consequence of the above disadvantages, especially of high variance.
- A sufficiently large tree is unbiased, as it can perfectly fit the train data, but is likely to due poorly on test data.

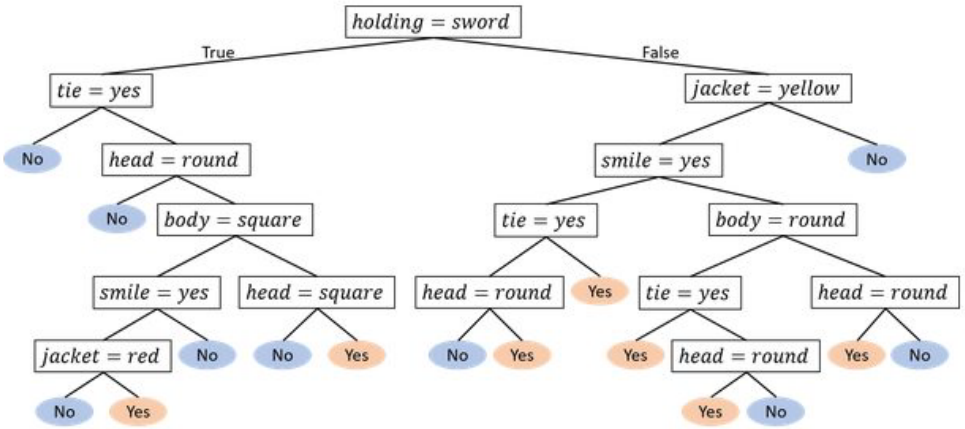
Predictive performance

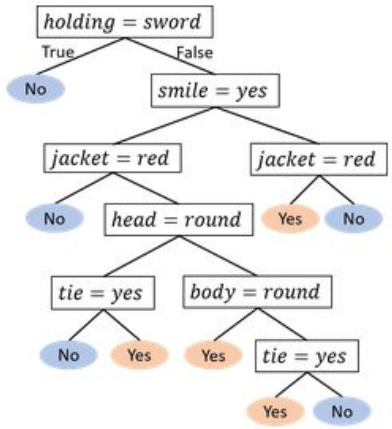
Ensembles of trees

- Can improve performance by combining multiple trees into an ensemble.
- Bagging averages many large trees to reduce variance.
- Boosting combines shallow trees to reduce their bias.
- Bagged and boosted trees typically the most accurate methods for structured data.
- However, they lack interpretability.

Beyond greedy methods

Monk 2: Six features





Left

- 16-leaf decision tree learned by CART.
- Training accuracy: 75.74%; test accuracy: 69.44%.

Right

- 9-leaf decision tree generated by GOSDT.
- Training accuracy: 81.07%; test accuracy: 73.15.

Optimal decision trees

- One does not know whether poor performance is due to the choice of model form (decision tree of a specific size) or poor optimization.
- In many problems, a fully optimized tree can be as accurate as an ensembles of trees.
- In other problems, getting the optimal tree largely discards the learning algorithm as the cause of poor performance, and may justify resorting to black-boxes.

Optimal decision trees

For a tree d, the objective function may be defined as a combination of prediction error and a sparsity penalty on the number of leaves:

$$R(d, \mathbf{X}, \mathbf{y}) = \ell(d, \mathbf{X}, \mathbf{y}) + \lambda |d|,$$

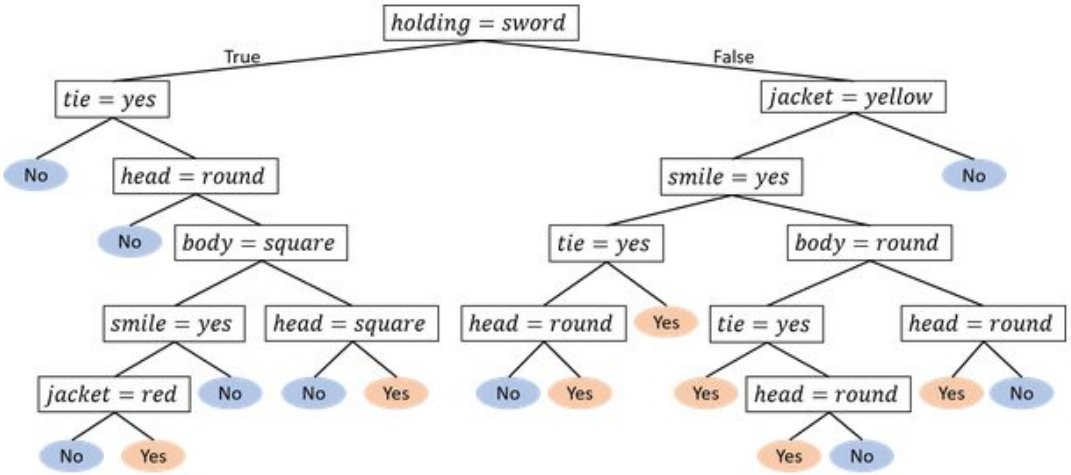
where the loss $\ell(d, \mathbf{X}, \mathbf{y})$ is the training set misclassification error of d and $|d|$ is the number of leaves in d.

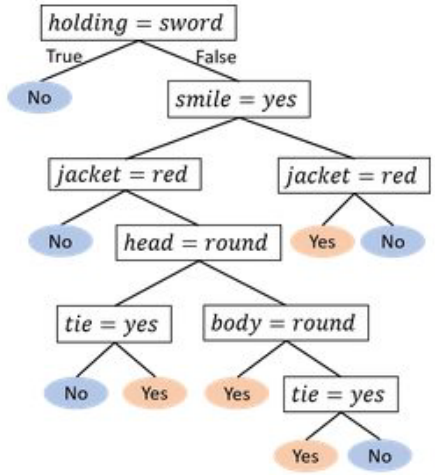
Optimal decision trees

Branch-and-bound

- Analytical bounds reduce the search space.
- Software engineering: adequate data structures (versus generic solvers).
- An algorithm: Globally optimal sparse decision trees (GOSDT).

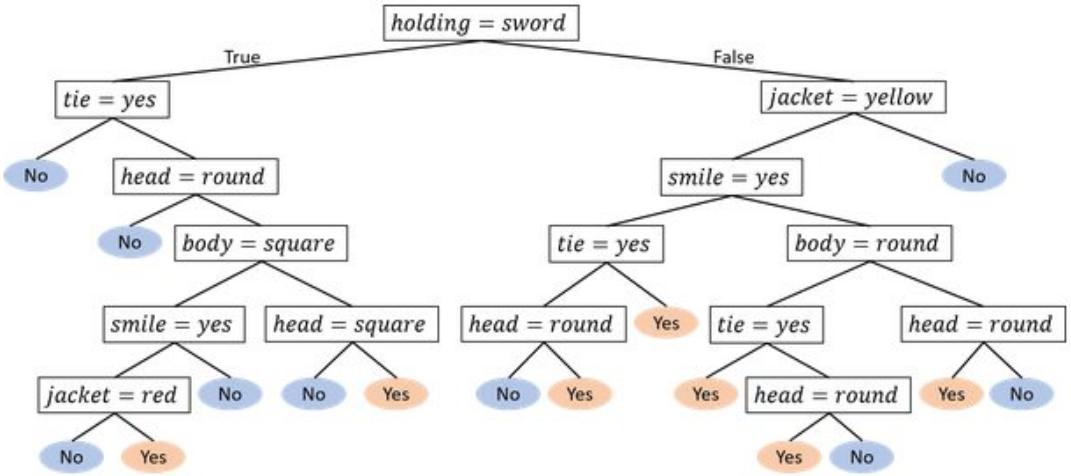
Optimal decision trees

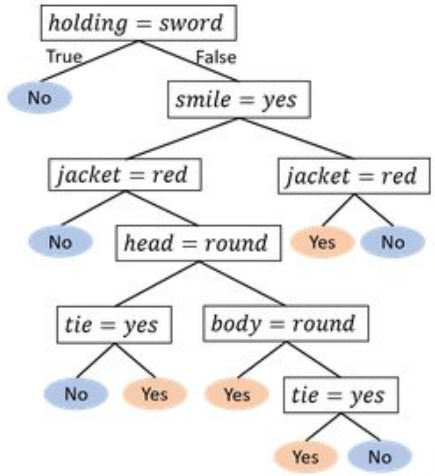




Note that, while the first split is the same, the second level already differs between the trees.

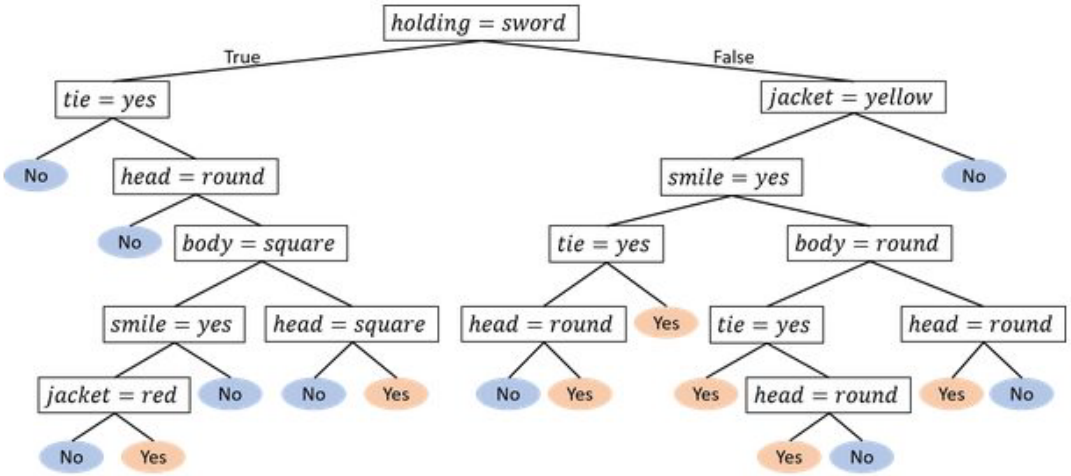
Optimal decision trees

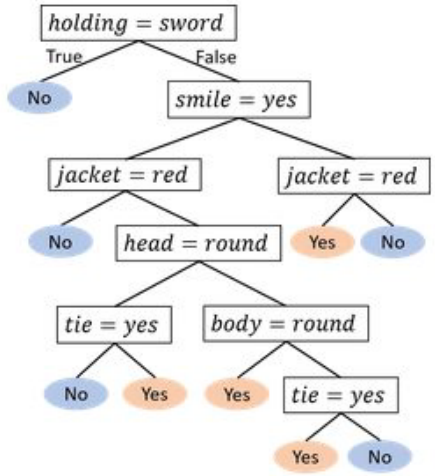




- Why doesn't the optimal tree branch further when `holding = sword` is true?

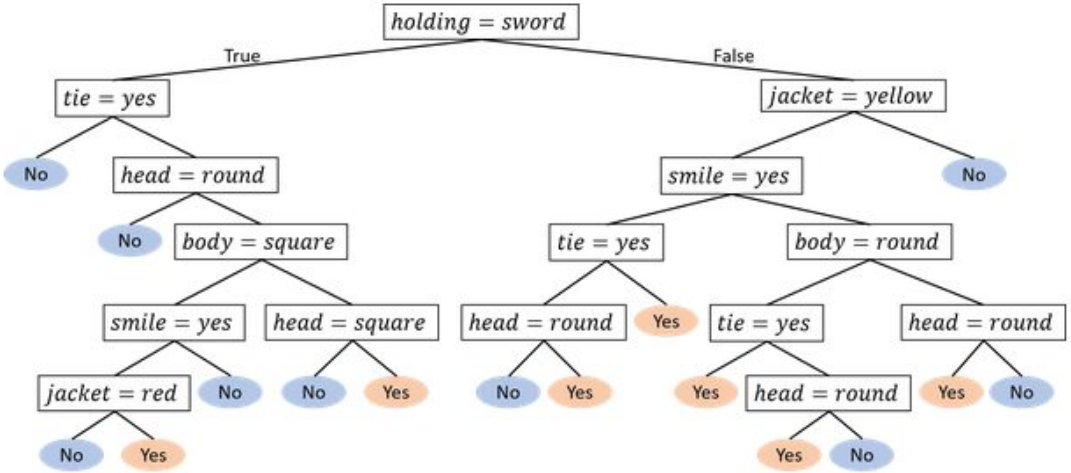
Optimal decision trees

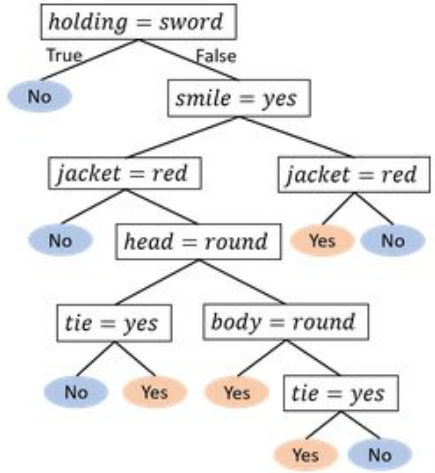




- Why doesn't the optimal tree branch further when `holding = sword` is true?
- The data points in the two `Yes` leaves under `holding = sword` will be misclassified in the training set, but the penalty due to additional leaves (6 in the CART tree) offsets the gains in accuracy.

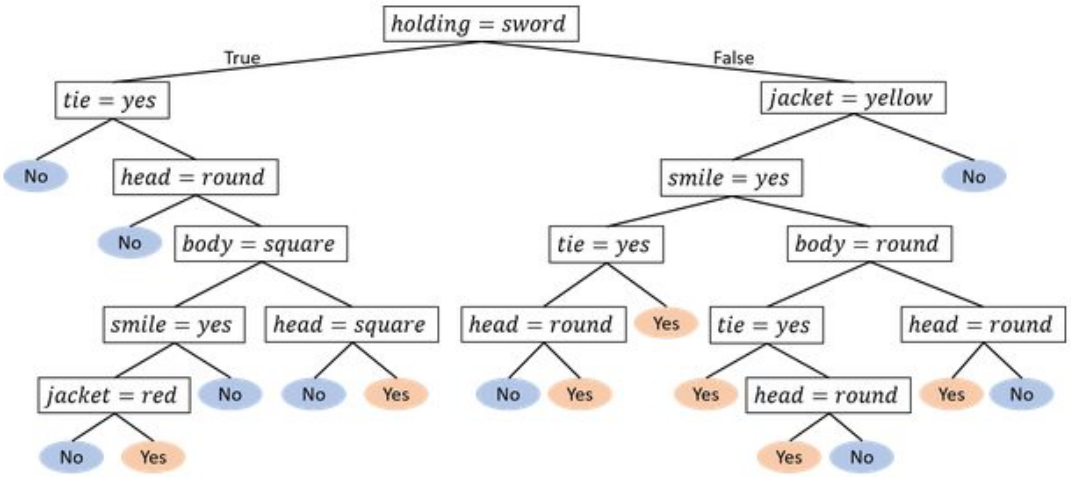
Optimal decision trees

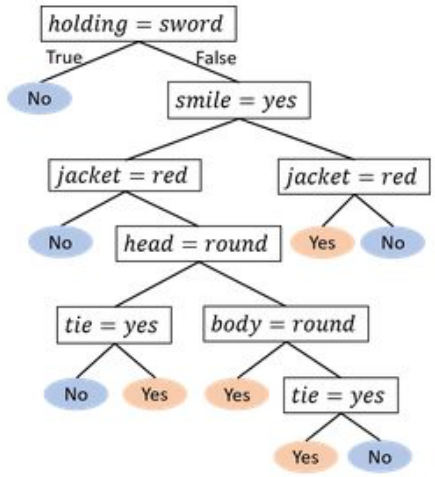




- Since we are optimizing on the training data, what would happen if we did not have a sparsity penalty?

Optimal decision trees





- Since we are optimizing on the training data, what would happen if we did not have a sparsity penalty?
- We would get a maximally grown tree (unless constraint such as minimum leaf size).

Note that the true classification "rule" in the Monk 2 problem is: exactly two of the six features have their first value. Limited training data: 169 out of 432 possible instances.

Optimal decision trees

Limitations

- Optimal decision tree methods can handle medium-sized datasets (tens of binary variables, thousands of samples) in reasonable time (e.g., within 10 minutes) with appropriate sparsity constraints.
- Scaling up to large datasets remains a challenge: Complexity exponential in n.
- Reducing this space through variable screening theorems or other means would be extremely helpful.

Decision lists and sets

- Each antecedent is a conjunction of attribute tests of the type `age>58` and `sex=male`, without an upper limit on their number.
- Possibly the most interpretable prediction models.

Example: Decision list

```
if ( age =18-20) and ( sex = male ) then predict yes
else if (age =21-23) and (priors =2-3) then predict yes
else if ( priors = >3 ) then predict yes
else predict no
```

Note that continuous features have been discretized; e.g., 21-23 is a category of the discretized age variable.

Decision lists and sets

Combining rules

- Mutually exclusive? What if two or more different rules give contradictory predictions?
- Exhaustive? What if no rule applies to an instance?

Questions

- Simple solution to ensure the rules are exhaustive?

Decision lists and sets

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- Simple solution to ensure the rules are exhaustive? Specify a default rule.
- A sensible prediction for the default rule is... ?

Decision lists and sets

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- Exhaustive? What if no rule applies to an instance?

Questions

- Simple solution to ensure the rules are exhaustive? Specify a default rule.
- A sensible prediction for the default rule is... ? The most frequent class among instances not covered by other rules.

Decision lists and sets

Two main strategies for ensuring, or dealing with the lack of, mutual exclusivity.

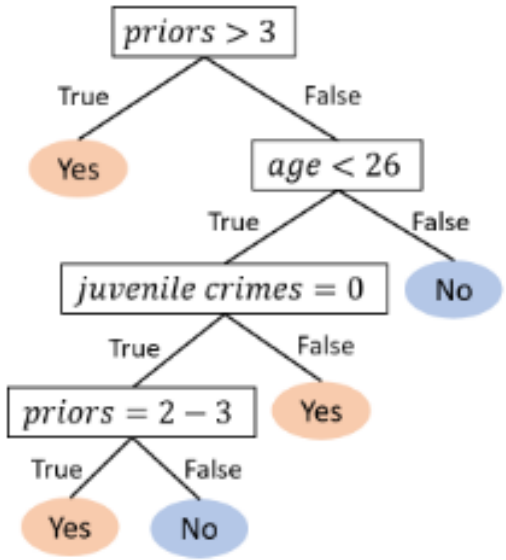
Decision list

- Ordered collection of rules.
- Allow for overlapping rules: return the prediction of first applicable rule.
- That is, the rules are tried in order, and the first that is satisfied makes the prediction.

Decision set

- Unordered collection of rules.
- Either ensure that the rules are mutually exclusive, ...
- ... or resolve conflicts by, e.g., majority voting.

Decision lists and sets



if $(age < 26)$ **and** $(priors = 2-3)$ **then predict** yes
else if $(juvenile\ crimes = 0)$ **and** $(priors < 3)$ **then predict** no
else predict yes

(b)

if $(priors > 3)$ **and** $(age < 21)$ **then predict** yes
if $(juvenile\ crimes > 0)$ **and** $(prior > 3)$ **then predict** yes
if $(age < 23)$ **and** $(prior = 2-3)$ **then predict** yes
else predict no

Tree to list/set

- Each path to R_m is a decision rule.
- Ordering is not an issue: the R_m are mutually exclusive.
- No need for a default rule: the R_m are exhaustive.
- Not compact: e.g., could replace all rules for the negative class with a default rule.
- Rules can often be further simplified, at the expense of overlapping.

Decision lists and sets



Decision lists and sets

Interpretability

- Lists: The latter rules are harder to interpret as they include the negation of previous rules.
- Sets: Reduced interpretability when several rules apply.

Decision lists and sets

Interpretability: Example

If Respiratory-Illness=Yes **and** Smoker=Yes **and** Age$\geq$ 50 **then** Lung Cancer

If Risk-LungCancer=Yes **and** Blood-Pressure$\geq$ 0.3 **then** Lung Cancer

If Risk-Depression=Yes **and** Past-Depression=Yes **then** Depression

If BMI$\geq$ 0.3 **and** Insurance=None **and** Blood-Pressure$\geq$ 0.2 **then** Depression

If Smoker=Yes **and** BMI$\geq$ 0.2 **and** Age$\geq$ 60 **then** Diabetes

If Risk-Diabetes=Yes **and** BMI$\geq$ 0.4 **and** Prob-Infections$\geq$ 0.2 **then** Diabetes

If Doctor-Visits $\geq$ 0.4 **and** Childhood-Obesity=Yes **then** Diabetes

If Respiratory-Illness=Yes **and** Smoker=Yes **and** Age$\geq$ 50 **then** Lung Cancer

Else if Risk-Depression=Yes **then** Depression

Else if BMI $\geq$ 0.2 **and** Age$\geq$ 60 **then** Diabetes

Else if Headaches=Yes **and** Dizziness=Yes, **then** Depression

Else if Doctor-Visits$\geq$ 0.3 **then** Diabetes

Else if Disposition-Tiredness=Yes **then** Depression

Else Diabetes

A decision set (left) and a decision list (right) learned from the same medical diagnosis dataset.

Decision lists and sets

Interpretability: Example

If Respiratory-Illness=Yes **and** Smoker=Yes **and** Age$\geq$ 50 **then** Lung Cancer

If Risk-LungCancer=Yes **and** Blood-Pressure$\geq$ 0.3 **then** Lung Cancer

If Risk-Depression=Yes **and** Past-Depression=Yes **then** Depression

If BMI$\geq$ 0.3 **and** Insurance=None **and** Blood-Pressure$\geq$ 0.2 **then** Depression

If Smoker=Yes **and** BMI$\geq$ 0.2 **and** Age$\geq$ 60 **then** Diabetes

If Risk-Diabetes=Yes **and** BMI$\geq$ 0.4 **and** Prob-Infections$\geq$ 0.2 **then** Diabetes

If Doctor-Visits $\geq$ 0.4 **and** Childhood-Obesity=Yes **then** Diabetes

If Respiratory-Illness=Yes **and** Smoker=Yes **and** Age$\geq$ 50 **then** Lung Cancer

Else if Risk-Depression=Yes **then** Depression

Else if BMI $\geq$ 0.2 **and** Age$\geq$ 60 **then** Diabetes

Else if Headaches=Yes **and** Dizziness=Yes, **then** Depression

Else if Doctor-Visits$\geq$ 0.3 **then** Diabetes

Else if Disposition-Tiredness=Yes **then** Depression

Else Diabetes

The 3rd rule in the list appears to be simple:

If BMI $\geq$ 0.2 and Age $\geq$ 60 then Diabetes,

but it actually is:

If not (Respiratory-Illness=Yes and Smoker=Yes and Age $\geq$ 50) and not (Risk-Depression=Yes) and BMI $\geq$ 0.2 and Age $\geq$ 60, then Diabetes.

Decision lists and sets

Interpretability: Example

If Respiratory-Illness=Yes **and** Smoker=Yes **and** Age$\geq$ 50 **then** Lung Cancer

If Risk-LungCancer=Yes **and** Blood-Pressure$\geq$ 0.3 **then** Lung Cancer

If Risk-Depression=Yes **and** Past-Depression=Yes **then** Depression

If BMI$\geq$ 0.3 **and** Insurance=None **and** Blood-Pressure$\geq$ 0.2 **then** Depression

If Smoker=Yes **and** BMI$\geq$ 0.2 **and** Age$\geq$ 60 **then** Diabetes

If Risk-Diabetes=Yes **and** BMI$\geq$ 0.4 **and** Prob-Infections$\geq$ 0.2 **then** Diabetes

If Doctor-Visits $\geq$ 0.4 **and** Childhood-Obesity=Yes **then** Diabetes

If Respiratory-Illness=Yes **and** Smoker=Yes **and** Age$\geq$ 50 **then** Lung Cancer

Else if Risk-Depression=Yes **then** Depression

Else if BMI $\geq$ 0.2 **and** Age$\geq$ 60 **then** Diabetes

Else if Headaches=Yes **and** Dizziness=Yes, **then** Depression

Else if Doctor-Visits$\geq$ 0.3 **then** Diabetes

Else if Disposition-Tiredness=Yes **then** Depression

Else Diabetes

- Reasoning about a rule in list requires reasoning about all of the preceding rules as well.
- Rules in decision sets are self-contained and apply individually, and so no complicated reasoning about multiple rules is needed.

Decision lists and sets

Learning

- For lists we need not worry about the rules being mutually exclusive.
- On the other hand, we need to consider the ordering of the rules; we would want the more accurate applicable rule to be triggered first.

Decision lists and sets

Example: Falling decision lists

- Probability of malignancy (positive class) decreases down the list.
- Healthcare applications: highest at-risk patients often considered first.
- Simplified interpretation for high-risk patients: only few rules need be considered for them.

Conditions			Probability	Support
IF	IrregularShape AND Age $\geq$ 60	THEN malignancy risk is	85.22%	230
ELSE IF	SpiculatedMargin AND Age $\geq$ 45	THEN malignancy risk is	78.13%	64
ELSE IF	IllDefinedMargin AND Age $\geq$ 60	THEN malignancy risk is	69.23%	39
ELSE IF	IrregularShape	THEN malignancy risk is	63.40%	153
ELSE IF	LobularShape AND Density $\geq$ 2	THEN malignancy risk is	39.68%	63
ELSE IF	RoundShape AND Age $\geq$ 60	THEN malignancy risk is	26.09%	46
ELSE		THEN malignancy risk is	10.38%	366

Decision lists and sets versus trees

Advantages

- More compact while equally expressive. Trees can have replicated sub-trees: left- and right-sub-tree can have the same structure.
- Furthermore, decision sets with overlapping rules can be more expressive than decision trees.

Disadvantages

Most implementations do not handle:

- Regression (only Classification).
- Continuous features (only categorical ones).

Decision lists: Learning

- Typically with greedy algorithms such as sequential covering.
- A well known instance of sequential covering is RIPPER.

Rules: accuracy versus support

- High support: applies to a large fraction of the instances.
- Accurate: correctly predicts the class for most of its support.
- By adding more attribute tests to the antecedent we can increase its accuracy while decreasing support.

		Accuracy	Support
IF petal length < 2.5	THEN predict setosa.	100%	33%
IF petal length ≥ 2.5 AND petal width < 1.8	THEN predict versicolor.	91%	36%
IF petal length ≥ 2.5 AND petal width ≥ 1.8	THEN predict virginica.	98%	31%

Decision lists: Learning

Sequential covering outline (binary classification)

1. Let L = empty decision list.
2. While L below a quality threshold or not all positive examples covered:
   1. Learn a rule r.
   2. Add rule r to L.
   3. Remove all data points covered by r.
   4. Go back to 2. with the remaining data.
3. Return the decision list L.

Decision lists: Learning

- Learning an optimal rule is computationally expensive.
- Typically we use a greedy procedure: grow and then prune a rule.

Simple algorithm: OneR

- Uses all values of a single feature.
- A single-node decision tree - yet not necessarily binary.

Decision lists: Modern methods

Bayesian rule lists

- Start from pre-mined frequent patterns in the data.
- Use Bayesian statistics to estimate and chose from a distribution over decision lists.

Certifiably Optimal Rule Lists for Categorical Data

- Takes pre-mined antecedents as input, not the original data.

Challenges

- Scale to many features.
- Handle continuous features.

Certifiably Optimal Rule Lists for Categorical Data

Frequent patterns

```
age>25,likes_pie=true,lives_west=true,drives=true,uses_windows=true
1,0,0,1,0
0,1,1,1,0
1,1,1,0,1
0,0,1,1,1
```

Limitations

- Binary classification.
- At most two clauses per rule.

https://corels.eecs.harvard.edu/corels/